

Game Design Document

Echoes of Aether

Version: 1.0

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Engine: Unity 6

Genre: JRPG / Turn-Based RPG

Platform: Windows PC

Target Audience: RPG dan JRPG Players

Executive Summary

Echoes of Aether adalah game 3D Narrative RPG dengan eksplorasi side-perspective pada lingkungan 3D dan pertarungan turn-based.

Project ini dirancang dengan pendekatan **data-driven architecture** menggunakan Scriptable Object sehingga konten baru dapat ditambahkan tanpa perlu mengubah kode inti.

Versi MVP berfungsi sebagai vertical slice yang menunjukkan seluruh gameplay loop utama, sementara arsitekturnya memungkinkan pengembangan menjadi RPG yang lebih besar.

Project Goals

Primary Goal

Membangun framework RPG yang scalable dan maintainable.

Secondary Goal

Memenuhi seluruh requirement technical test:

- Player Movement

- NPC Interaction
- Dialogue System
- Cutscene System
- Turn-Based Battle
- Prefab Workflow
- Scriptable Object Architecture

Long-Term Goal

Menjadi foundation untuk portfolio project dan pengembangan game komersial.

Core Pillars

Pillar 1 – Exploration

Player harus merasa sedang menjelajahi dunia yang hidup.

Fitur:

- Movement
 - NPC Interaction
 - Environmental Storytelling
-

Pillar 2 – Narrative

Cerita disampaikan melalui:

- NPC Dialogue
 - Cutscene
 - Event Trigger
-

Pillar 3 – Tactical Combat

Combat berfokus pada:

- Turn Order
 - Skill Selection
 - Resource Management
-

Pillar 4 – Extensibility

Seluruh sistem dibuat modular dan data-driven.

Core Gameplay Loop

```
Explore
↓
Talk To NPC
↓
Receive Objective
↓
Travel
↓
Event Trigger
↓
Battle
↓
Reward
↓
Progress Story
```

Player Systems

Movement System

Features

- 8 Direction Movement
- Input Buffering
- Animation State

Input

Input	Action
WASD	Move
Arrow Key	Move
Space	Interact
Esc	Pause

World Interaction System

Interaction Types

NPC

Memberikan:

- Dialogue
- Quest
- Event Trigger

Object

Contoh:

- Chest
- Sign Board
- Lever

Dialogue System

Functional Requirements

FR-DLG-001

Menampilkan dialog berdasarkan data.

FR-DLG-002

Mendukung multi-line dialogue.

FR-DLG-003

Menampilkan portrait speaker.

FR-DLG-004

Mendukung branching dialogue (future).

Dialogue Flow

```
Player Interact
↓
Dialogue Opens
↓
Line Displayed
↓
Player Continue
↓
Next Line
↓
Dialogue Ends
```

Cutscene System

Purpose

Mengontrol sequence tanpa input player.

Supported Commands

MVP

- Move Actor
- Wait
- Show Dialogue
- Play Animation

- Camera Focus
-

Future

- Timeline Integration
- Audio Cue
- Screen Fade

Combat Design

Combat Type

Turn-Based JRPG

Battle States

```
BattleStart  
PlayerTurn  
EnemyTurn  
Victory  
Defeat  
BattleEnd
```

Turn Sequence

```
Battle Start  
  
↓  
  
Determine Initiative  
  
↓  
  
Player Turn  
  
↓
```

Enemy Turn

↓

Repeat

↓

Victory / Defeat

Character System

Base Stats

Stat	Description
HP	Health
MP	Mana
ATK	Physical Damage
DEF	Damage Reduction
SPD	Initiative
LUK	Critical Chance

Derived Stats

Critical Chance

$$\text{CritChance} = \text{LUK} * 0.5\%$$

Damage

$$\text{Damage} = \text{ATK} - \text{DEF}$$

Skill System

Skill Categories

Physical

Contoh:

- Slash
 - Heavy Strike
-

Magic

Contoh:

- Fireball
 - Ice Spear
-

Support

Contoh:

- Heal
 - Protect
-

Skill Formula

```
Final Damage  
  
=  
  
(Base Power + User Attack)  
  
-  
  
Target Defense
```

Enemy AI

Behavior Tree Lite

Example

Enemy Turn

↓

HP < 30%

↓

Yes → Heal

↓

No

↓

Attack

Progression System

MVP

Tidak ada leveling.

Future

EXP

Battle

↓

Gain EXP

↓

Level Up

↓

Stat Increase

Quest System

Quest Structure

```
Quest
├ ID
├ Name
├ Description
├ Objectives
├ Rewards
└ State
```

Quest State

```
Inactive
Active
Completed
Failed
```

Save System

Future Scope

Data yang disimpan:

```
Player Position
Quest Progress
Inventory
Story Flags
Character Stats
```

Content Pipeline

Character Creation Flow

```
CharacterData SO
↓
Prefab
↓
Spawn
↓
Runtime Instance
```

Skill Creation Flow

```
SkillData SO
↓
Assign To Character
↓
Battle Usage
```

TECHNICAL DESIGN DOCUMENT (TDD)

17. Architecture Overview

Menggunakan pendekatan:

```
Data Layer
↓
Logic Layer
↓
Presentation Layer
```

Data Layer

Scriptable Objects.

Contoh:

```
CharacterData
SkillData
DialogueData
QuestData
ItemData
```

Logic Layer

Manager dan Gameplay Systems.

Contoh:

```
GameManager
BattleManager
DialogueManager
QuestManager
```

Presentation Layer

```
UI
Animation
VFX
Audio
```

18. Folder Structure

```
Assets
|
├─ Art
|
├─ Audio
|
├─ Animations
|
├─ Prefabs
|
└─ Characters
```

```
|
|   | NPC
|   | Enemy
|   | UI
|
| ScriptableObjects
|
|   | Characters
|   | Skills
|   | Dialogue
|   | Quests
|   | Items
|
| Scenes
|
|   | Bootstrap
|   | Village
|   | Forest
|   | Battle
|
| Scripts
|
|   | Core
|   | Managers
|   | Characters
|   | Battle
|   | Dialogue
|   | Quest
|   | UI
|
| Resources
```

19. Core Managers

GameManager

Tanggung jawab:

- Global State
- Scene Transition

DialogueManager

Tanggung jawab:

- Dialogue Queue
 - UI Control
-

BattleManager

Tanggung jawab:

- Turn Flow
 - Battle State
-

QuestManager

Tanggung jawab:

- Quest Tracking
 - Objective Validation
-

20. Event Architecture

Menggunakan Event-Driven Pattern.

Contoh:

```
OnBattleStart  
OnBattleEnd  
  
OnQuestAccepted  
OnQuestCompleted  
  
OnDialogueStarted  
OnDialogueFinished
```

Keuntungan:

- Loose Coupling
 - Mudah di-maintain
 - Mudah ditambah fitur baru
-

21. State Machine Design

Player State

```
Idle  
Move  
Interact  
Cutscene  
Battle
```

Battle State

```
Initialize  
  
↓  
  
PlayerTurn  
  
↓  
  
EnemyTurn  
  
↓  
  
Victory  
  
↓  
  
Defeat
```

22. Scriptable Object Design

CharacterData

```
CharacterName  
MaxHP  
MaxMP  
Attack
```

Defense
Speed
Portrait
SkillList

SkillData

SkillName
Description
Power
ManaCost
TargetType
Animation
Icon

DialogueData

Speaker
Portrait
DialogueLines

23. Future Expansion Roadmap

Version 0.1

Technical Test Build

Version 0.2

Quest System

Version 0.3

Inventory System

Version 0.4

Status Effect System

Version 0.5

Boss Encounter

Version 0.6

Save Load

Version 1.0

Full Playable Demo